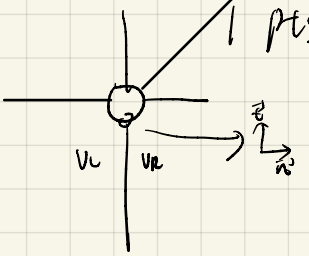
Section 2.4 (meshmotion.f)

We want get V_g for all vex in this file



1 pts has 4 edges $i = 1, \cdots 4$

$$u_t^i = \frac{V_L + V_R}{2}$$

$$u_n^i = u_s^i \quad W^i = u_s^i n + u_t^i t$$

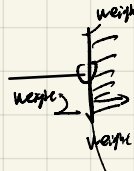
$$\text{interior} = \frac{1}{4} \sum W^i$$

Shu only discuss the interior mesh

We have to deal with boundary mesh pts

① periods same as interior

② wall (fix) → our own way!



can only move along the wall!

$W = W - (W \cdot n_w) n_w$ just set 0 on n_w.

We need find wall! the do calculation

Function compute_vertex_velocity

We need: **State** ↓ find all vex's Coord

Discrete_case ↓ problem, mesh, option, gamma

Step 1 for all face find wall → wall

for face in mesh.faces_ls

if right == 0, boundary => wall

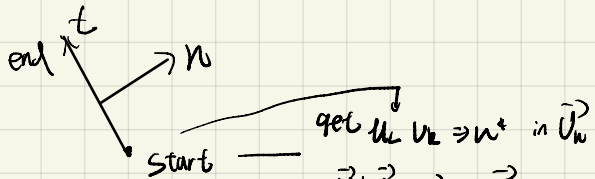
Because all our left face is good! by construction

(We update and moving vex in group level, because physically they are save vex but has two represent in mesh (period)

So once we find the vex is wall index, we take the whole group

mesh.vex2group_index[start, end] → wall_group_flag_ls = T

Step 2 for each face, we need tangle vector normal vector



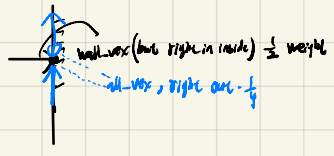
get $u_1, u_2 \Rightarrow u^2$ in $\vec{U_n}$

$\frac{\vec{U_1}+\vec{U_2}}{2} \cdot \vec{t} = \vec{U_0}$

get $U_t = \vec{U_0} \cdot \vec{t} + \vec{U_n} \cdot \vec{n}$ for start and end

---

if is





Well if wall but has r, l we give double weight.

other wise 1 weight

So each group we add U_i^{acx} for all possible i

and $U = \frac{\sum U_i^{acx}}{\sum \text{weight}^{acx}}$

Finally for all vex

$$U_{vex} = U_{vex} - (U_{vex} \cdot \vec{n}_{wall}) \cdot \vec{n}_{wall}$$

we can do it more than once, it's still the same

so we loop all face, we have $\vec{n}$ as $\vec{n}_{wall}$ if region=wall.

So For all vex of mesh

We know have its speed W_{vex}

Thus $\forall\ x \in$ segment (x_1, x_2)

$$u_g = u_1 \frac{x - x_2}{x_1 - x_2} + u_2 \frac{x - x_1}{x_2 - x_1}, \quad v_g = v_1 \frac{y - y_2}{y_1 - y_2} + v_2 \frac{y - y_1}{y_2 - y_1}, \tag{2.18}$$

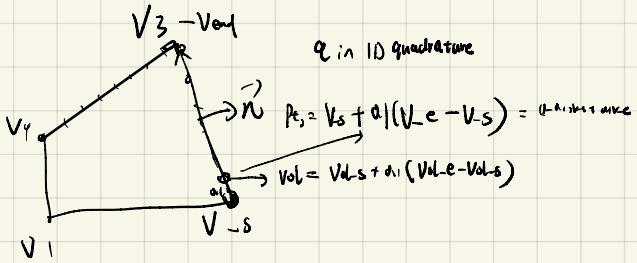
Basis motion

$$\xi = \frac{2(x-x_c)}{h_x}, \quad \eta = \frac{2(y-y_c)}{h_y}, \quad \cdots \quad K_b \xi^{b_i} \eta^{b_j} = \psi_b$$

$\phi_b = \psi_b - \bar{\psi}_b$, if we want to get $\frac{\partial \phi_b}{\partial t}$

we just need $\frac{\partial x_c}{\partial t}$ $\frac{\partial h_x}{\partial t}$ $\frac{\partial h_y}{\partial t}$ $\frac{\partial y_c}{\partial t}$

$$\frac{\partial (\text{area}_K)}{\partial t} = \oint_{\partial K} \vec{u}_g \cdot \vec{n} \, ds$$



$$x_c = \frac{\int_K x \cdot dA}{\text{area}_K} \qquad x_c \cdot \text{area}_K = \int_K x \, dA$$

$$\frac{\partial x_c}{\partial t} \cdot \text{area}_K + x_c \frac{\partial \text{area}_K}{\partial t} = \frac{\partial \int_K x \cdot dA}{\partial t}$$

$$\frac{\partial x_c}{\partial t} \text{area}_K + x_c \oint_{\partial K} \vec{u}_g \cdot \vec{n} = \int_{K(t)} \frac{\partial x}{\partial t} dA + \oint_{\partial K} x \, \vec{u}_g \cdot \vec{n} \, ds$$

($\int_{K(t)} \frac{\partial x}{\partial t} dA = 0$)

$$= \oint_{\partial K} x \, \vec{u}_g \cdot \vec{n} \, ds$$

(1) $$\frac{\partial x_c}{\partial t} = \frac{\oint_{\partial K} (x - x_c) \vec{u}_g \cdot \vec{n} \, ds}{\text{area}} \quad \text{1D} \quad = \dot{x}_c$$

Similarly (2) $$\frac{\partial x_c}{\partial t} = \frac{\oint_{\partial K} (y - y_c) \vec{u}_g \cdot \vec{n} \, ds}{\text{area}} \quad \text{1D} \quad = \dot{y}_c$$

$h_x = \text{Vex}_{x,\max} - \text{Vex}_{x,\min}$ | w_{Vex} mesh vertex speed we got it at last page

(3) $$\frac{\partial h_x}{\partial t} = w_{x,\max x} - w_{x,\min y} = \dot{h}_x$$

(4) $$\frac{\partial h_y}{\partial t} = w_{y,\max y} - w_{y,\min y} = \dot{h}_y$$

$$\partial_t \xi(x,y) = \frac{2h_x(-\dot{x}_c) - 2(x-x_c)\dot{h}_x}{(h_x)^2} = -\frac{2\dot{x}_c}{h_x} - \frac{2(x-x_c)}{h_x^2}\dot{h}_x$$

$$\partial_t \eta(x,y) = -\frac{2\dot{y}_c}{h_y} - \frac{2(y-y_c)}{h_y^2}\dot{h}_y$$

$$\dot{\psi}_i = \partial_t \psi_i = \partial_t(k_i \xi^a \eta^b) = k_i(a\xi^{a-1}\eta^b \partial_t\xi + b\xi^a \eta^{b-1}\partial_t\eta)$$

$$\dot{\bar{\psi}}_i = \partial_t \bar{\psi}_i = \partial_t \left(\frac{\int_K k_i \xi^a \eta^b \, dA}{\text{area}_K} \right)$$

∂_t mean_can_be_in

Reynold's transport theorem $= \left(\frac{\int_K \partial_t \psi_i + \oint_{\partial K} \psi_i \, u_g \cdot \vec{n} \, dS}{\text{area}_K} \right) - \frac{\int_K \psi_i}{\text{area}_K^2} \cdot \text{area}_K$

mean dt_can_be

To eval a single pt's $\dot{\psi}_i$ we need to calculate while $\dot{\psi}_i$ at all quadrature pt! (Volumn, quadrature)

I can split $u_g \cdot \vec{n}$ at quadrature edges as a new function

we write functions ① $\dot{\psi}_i$ at a pt

② $\dot{\psi}_i$ at all quadrature pt mat

$\partial_t \phi_i = \dot{\psi}_i - \dot{\bar{\psi}}_i$ ③ ϕ_i at a pt

④ ϕ_i at all quadrature pts mat

To make it more efficient I will build a structure

Struct cellMotionCache:
- cell_velocities_mat
- quadrature_coords_at_edge
- weights
- u_g_quadrature
- norm_quadrature
- dt_area
- dt_x_L
- dt_y_L
- dt_h_x
- dt_h_y
- dt_raw_basis_at_quadrature
- dt_mean_raw_basis
- dt_basis_at_quadrature

$$\frac{\cdot}{\psi_l} = \frac{\left(\int_K \partial_t \psi_l + \oint_{\partial K} \psi_l \, u_g \cdot \vec{n} \, dS\right)}{area_K} - \frac{\int_K \psi_b}{area_K^2} \cdot area_K$$

$$\frac{\int_K \partial_t \psi_l}{area_K} + \frac{\oint_{\partial K} \psi_l \, u_g \cdot \vec{n} \, dS}{area_K}$$

↓ dt_raw_basis_mat

$$\begin{bmatrix} \partial_t \varphi_1(p_1) & \cdots & \partial_t \varphi_1(p_n) \\ & \vdots & \\ \partial_t \varphi_l(p_1) & & \end{bmatrix} \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} = \begin{pmatrix} \sum \partial_t \varphi(p_i) w_i \\ \vdots \\ \sum \partial_t \varphi_{(l)}(p_i) w_i \end{pmatrix}$$

$\Sigma \psi_l(p_i) \cdot \vec{u_g} \cdot \vec{n} \cdot w_i$

↓ dt_raw_basis_mean

↓ boundary_flux_raw_basis_mean

$$- \frac{raw_basis_mean}{area} \cdot dt_area$$